

AUSTIN, MN, USA

WMO: 727566

Lat: 43.665N

Lon: 92.933W

Elev: 376

StdP: 96.89

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04902

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.1	-22.1	-28.0	0.3	-23.4	-26.1	0.4	-21.8	12.4	-8.6	11.4	-8.8	3.3	300	0.556

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	32.0	23.6	30.0	22.6	28.0	21.5	25.5	29.4	24.2	28.2	23.2	27.0	5.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.8	27.9	22.8	18.3	26.8	22.1	17.6	26.2	80.4	29.6	75.1	27.8	70.7	27.0	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.9	9.3	8.3	-28.2	34.2	3.4	1.8	-30.6	35.5	-32.6	36.6	-34.5	37.6	-36.9	38.9		
(5)			-28.3	27.0	3.2	1.6	-30.6	28.1	-32.5	29.1	-34.3	30.0	-36.6	31.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.7	-9.1	-6.9	0.2	7.8	14.5	20.1	22.3	20.6	16.8	9.6	1.6	-5.7
	DBStd	12.22	7.37	7.02	6.69	5.55	4.75	3.62	2.96	2.98	4.62	5.14	6.12	6.56
	HDD10.0	2317	592	474	312	104	13	0	0	0	4	72	259	486
	HDD18.3	4220	850	708	562	318	138	23	4	12	83	275	503	744
	CDD10.0	1483	0	0	8	38	152	303	380	330	208	58	7	0
	CDD18.3	345	0	0	0	2	18	76	126	83	37	3	0	0
	CDH23.3	2899	0	0	1	27	187	630	1081	620	315	38	1	0
	CDH26.7	818	0	0	0	6	56	182	337	151	79	8	0	0
(14) Wind	WSAvg	3.7	4.2	4.2	4.2	4.5	4.0	3.3	2.8	2.5	3.1	3.8	4.0	3.9
(15) Precipitation	PrecAvg	911	24	22	51	94	111	136	124	124	86	60	47	32
	PrecMax	1099	52	51	88	185	202	237	285	297	243	195	151	80
	PrecMin	646	6	4	8	27	46	57	28	32	25	14	3	4
	PrecStd	127	13	13	25	45	43	48	57	58	56	41	38	20
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.5	11.1	21.4	27.2	32.0	32.9	34.0	32.3	32.0	27.6	20.0	12.2
		MCWB	3.9	8.1	14.7	16.6	20.8	23.6	25.3	24.0	22.4	18.2	13.3	9.6
	2%	DB	3.8	7.1	16.7	22.5	27.7	31.0	32.2	29.9	28.8	23.0	16.3	7.3
		MCWB	1.7	5.0	12.6	13.9	18.1	22.3	24.8	23.6	21.5	16.0	11.4	5.1
	5%	DB	2.3	3.9	12.7	19.1	25.1	28.7	30.1	27.9	27.2	20.9	12.8	4.1
		MCWB	0.8	2.2	9.4	12.2	17.1	20.9	23.5	22.4	20.1	15.4	9.1	2.4
	10%	DB	1.0	2.4	9.0	17.2	22.6	27.2	28.0	27.1	24.9	17.7	10.2	2.5
		MCWB	0.0	1.1	6.7	10.9	15.8	20.1	22.2	21.6	19.0	13.2	7.1	1.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.2	8.5	16.3	17.4	22.8	25.5	27.9	26.8	24.6	20.4	14.8	9.9
		MCDB	7.0	10.0	20.0	23.5	28.4	30.3	31.4	29.6	28.5	24.7	18.5	12.2
	2%	WB	2.1	4.9	13.1	15.2	20.2	23.8	25.8	24.9	22.9	17.9	12.1	5.2
		MCDB	3.7	6.8	16.0	20.6	25.5	28.5	29.7	28.3	27.1	20.7	15.3	7.0
	5%	WB	0.8	2.7	9.8	13.4	18.7	22.6	24.6	23.6	21.4	15.8	9.6	2.9
		MCDB	1.8	3.8	12.9	18.4	23.3	27.2	28.7	27.0	25.4	19.6	13.0	4.2
	10%	WB	-0.4	1.0	6.7	11.6	17.1	21.3	23.5	22.5	19.9	13.7	7.5	1.4
		MCDB	0.6	2.3	9.1	16.0	21.0	25.8	27.4	25.7	23.9	17.5	9.7	2.4
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	8.5	8.6	10.4	10.7	10.2	10.4	10.3	11.4	10.3	8.5	7.5
		MCDBR	9.4	9.0	13.0	15.4	13.9	12.4	11.8	11.4	13.2	14.5	12.6	9.4
	5% WB	MCWBR	8.0	7.7	9.0	8.8	7.0	6.0	6.1	6.4	7.0	8.1	8.6	7.9
		MCWBR	8.6	8.4	12.6	13.4	11.4	10.7	10.5	10.0	11.3	11.6	11.7	8.5
(40) Clear-Sky Solar Irradiance	taub		0.266	0.291	0.314	0.365	0.392	0.410	0.415	0.403	0.371	0.326	0.289	0.265
		taud	2.418	2.341	2.410	2.315	2.300	2.289	2.269	2.305	2.366	2.483	2.506	2.448
	Ebn at Noon		877	905	928	898	880	862	853	851	853	857	844	845
		Edh at Noon	74	95	102	122	128	130	131	122	106	83	68	65
(44) All-Sky Solar Radiation	RadAvg		1.77	2.83	3.75	4.61	5.42	6.05	6.31	5.33	4.26	2.77	1.82	1.34
	RadStd		0.19	0.30	0.38	0.48	0.41	0.46	0.28	0.37	0.30	0.37	0.16	0.11

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	-0.75	N/A	N/A	N/A	N/A	N/A
(47)	Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page